

3- ϕ transformers

$$v = V_m \sin \omega t ; i = I_m \sin(\omega t - \phi)$$

$$P_{(1-\phi)} = v \cdot i$$

$$= V_m \sin \omega t \cdot I_m \sin(\omega t - \phi)$$

$$= \frac{V_m I_m}{2} \cdot 2 \sin \omega t \cdot \sin(\omega t - \phi)$$

$$= -\frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} [\cos(\omega t - \omega t + \phi) - \cos(\omega t + \omega t - \phi)]$$

$$= V \cdot I [\cos \phi - \cos(2\omega t - \phi)]$$

$$P_{(1-\phi)} = V \cdot I \cos \phi - V I \cos(2\omega t - \phi)$$

$$P = \frac{dW}{dt} \quad \omega = qv$$

$$= v \frac{dq}{dt}$$

$$= v i$$

So, a 1- ϕ power suffers from double freq. component

3000 kVA

option 1: 1x3000kVA, 3- ϕ unit

option 2: 3x1000 kVA, 1- ϕ TF connected in 3- ϕ bank

POLARITY TEST

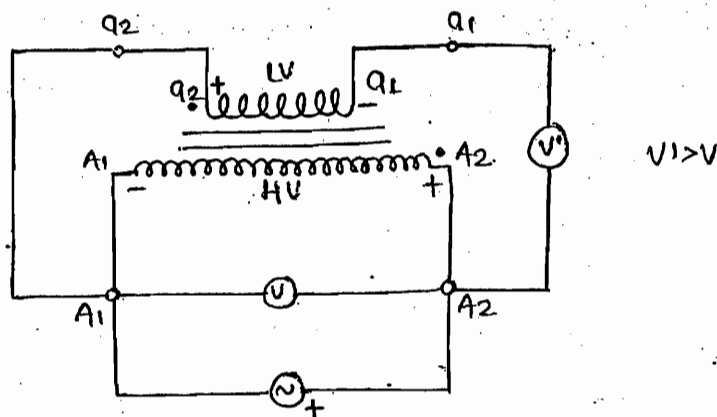


Fig:- additive polarity

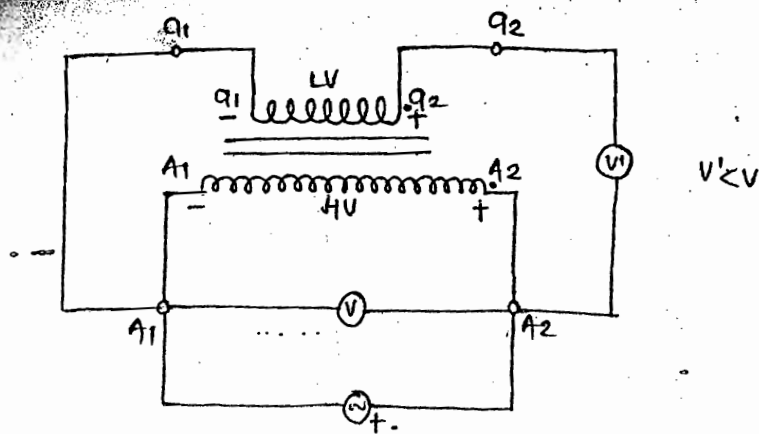


Fig:- Subtractive polarity.

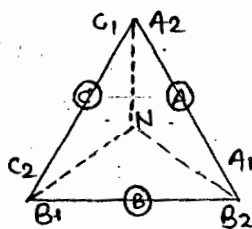
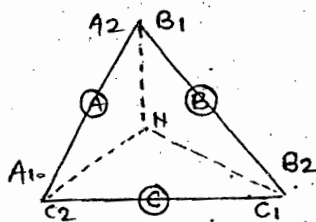
* Phasor Groups →

Group (1) :- $0^\circ \rightarrow Yy0, Dd0, Dz0$

Group (2) :- $180^\circ \rightarrow Yy6, Dd6, Dz6$

Group (3) :- $30^\circ (\text{lag}) \rightarrow Yd1, Dy1, Yz1$

Group (4) :- $30^\circ (\text{lead}) \rightarrow Yd11, Dy11, Yz11$

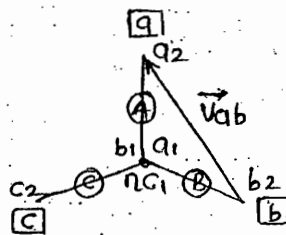
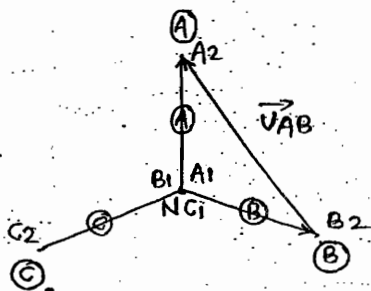
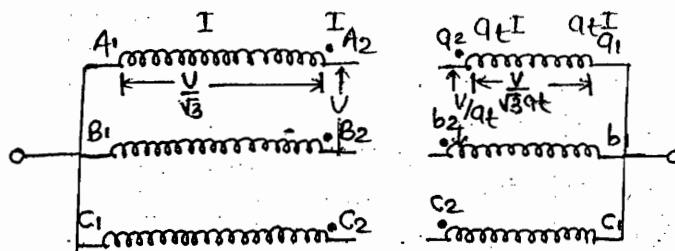


* British Practise for phasor groups →

- (1.) Phasor seq. in ABC.
- (2.) HV line (A-N) phasor always at 0 o'clock.
- (3.) HV line 'A' terminal to be taken from A_2
- (4.) $\vec{V}_{q_2 q_1}$ is in phase with $\vec{V}_{A_2 A_1}$.

Y40 →

$$q_t = \frac{N_{HV}}{N_{LV}}$$



$$V_{AB} = V_{AN} - V_{BN}$$

$$V_{AN} = V_{AB} + V_{BN}$$

$$\text{Phase vol. transformation Ratio} = \frac{V}{\sqrt{3}} : \frac{V}{\sqrt{3} q_t} = q_t : 1$$

$$\text{line vol. transformation Ratio} = V : \frac{V}{q_t} = q_t : 1$$

$$S_{HV} = \sqrt{3} V I$$

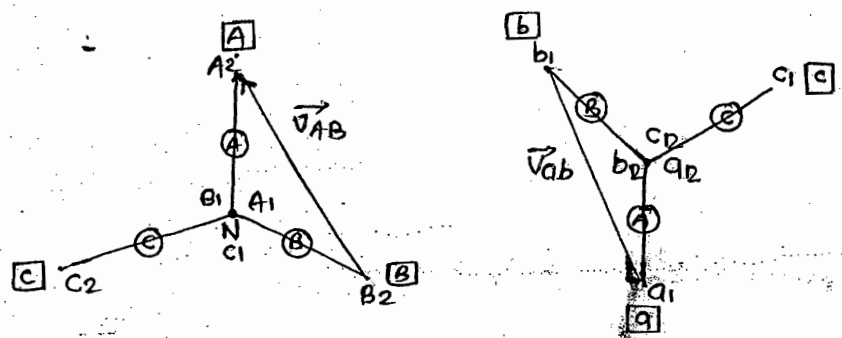
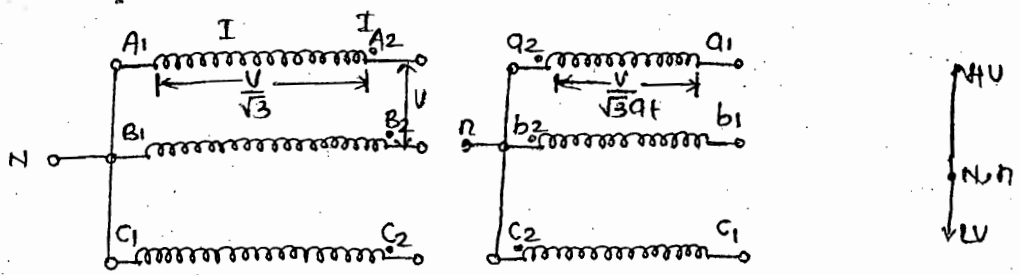
$$S_{LV} = \sqrt{3} \left(\frac{V}{q_t} \right) (q_t I)$$

$$= \sqrt{3} V I$$

$$= S_{HV}$$

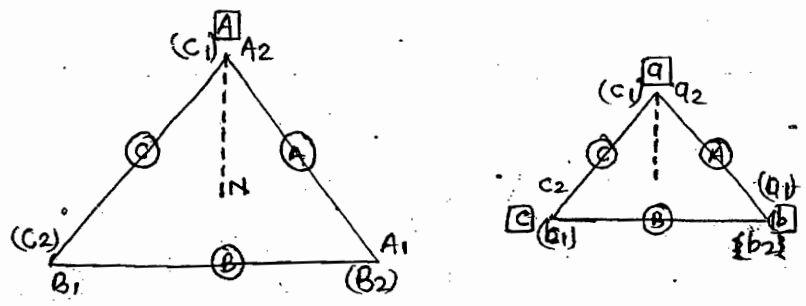
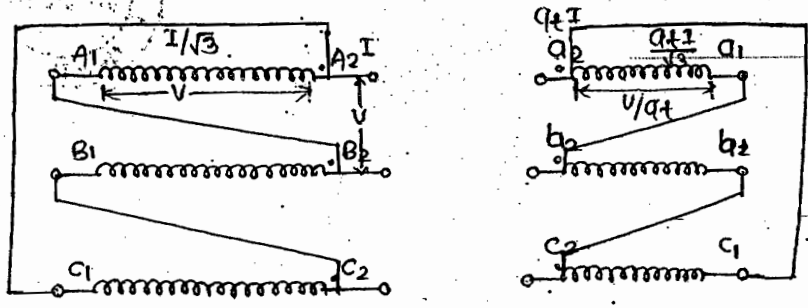
$$\boxed{S_{LV} = S_{HV}}$$

Yy6 →



Dd0 →

* The British Convention for Δ - Δ connection is that the HV side Δ & LV side Δ with the same combinations; this shows that a_1b_2, b_1c_2, c_1a_2



phase vol. X mation ratio = $V: \frac{V}{a_t} = a_t:1$

line vol. X mation ratio = $V: \frac{V}{a_t} = a_t:1$

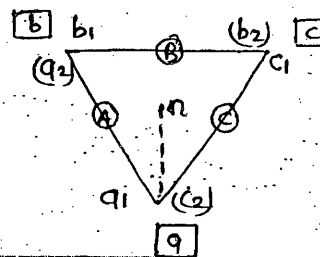
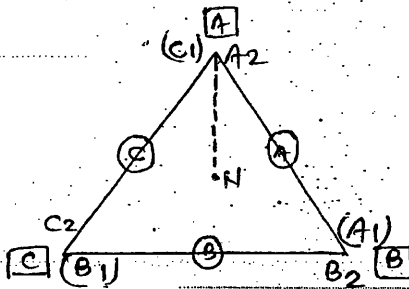
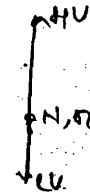
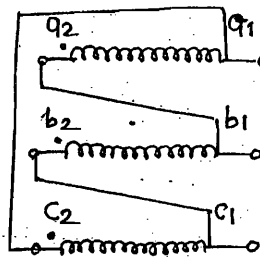
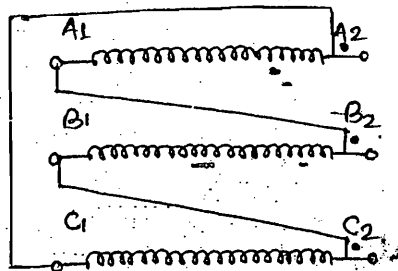
$S_{HV} = \sqrt{3}VI$

$S_{LV} = \sqrt{3} \left(\frac{V}{a_t} \right) (a_t I)$

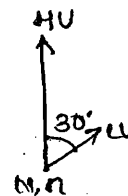
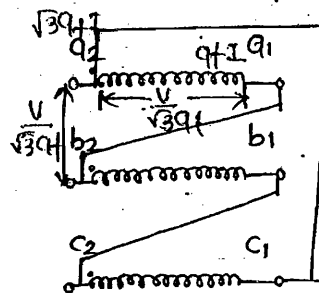
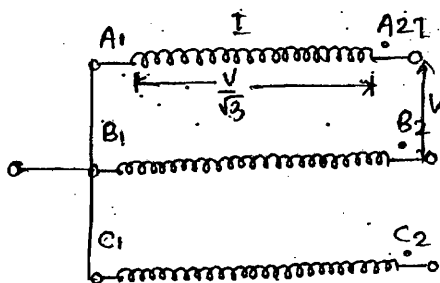
$= \sqrt{3}VI$

$S_{LV} = S_{HV}$

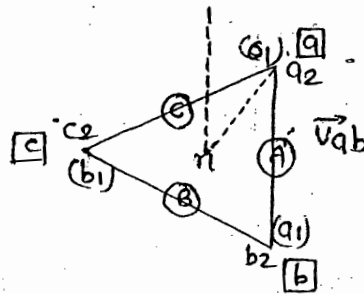
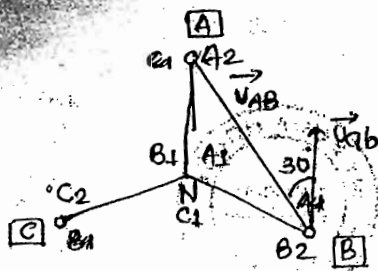
Dd6 →



Yd1 →



Sagar Sen
887145336



$$\text{Phase vol. X ratio} = \frac{V}{\sqrt{3}} : \frac{V}{\sqrt{3}q_t} = q_t : 1$$

$$\text{line vol. X ratio} = V : \frac{V}{\sqrt{3}q_t} = \sqrt{3}q_t : 1$$

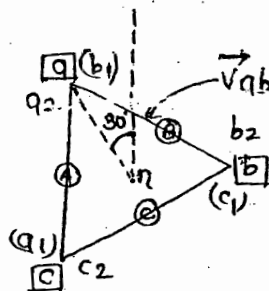
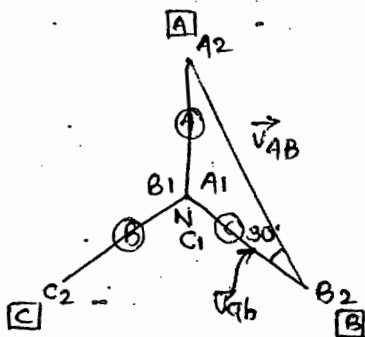
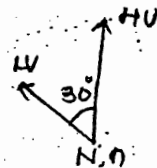
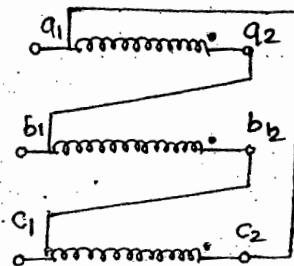
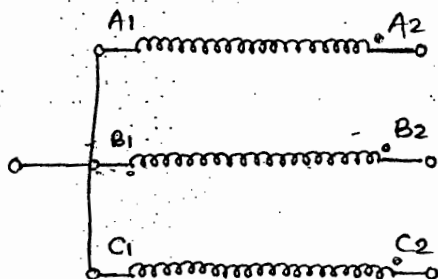
$$S_{HV} = \sqrt{3}VI$$

$$S_{LV} = \sqrt{3} \left(\frac{V}{\sqrt{3}q_t} \right) (\sqrt{3}q_t I)$$

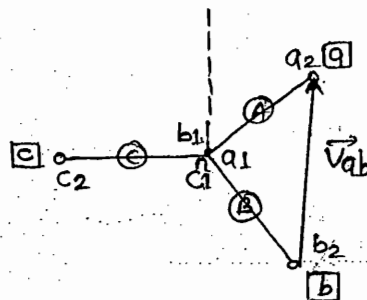
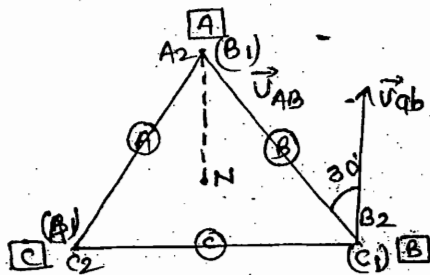
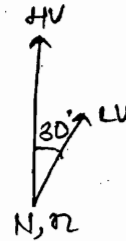
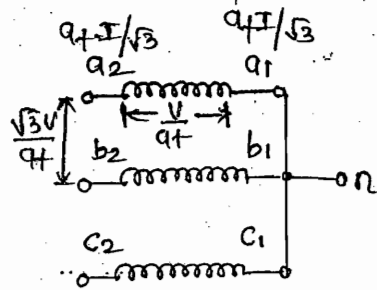
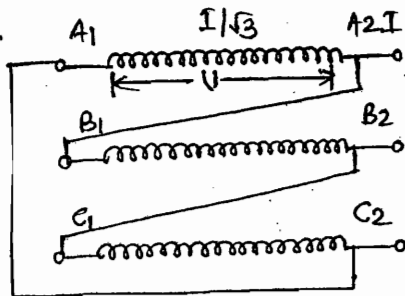
$$= \sqrt{3}VI$$

$$S_{LV} = S_{HV}$$

Yd11 →



Dy1 →



Hint → Δ limb is decided by λ

Phase vol. X transformation ratio = $V : \frac{V}{\sqrt{3}} = q_1 : 1$

line vol. X transformation ratio = $V : \frac{\sqrt{3}V}{\sqrt{3}} = \frac{q_1}{\sqrt{3}} : 1$

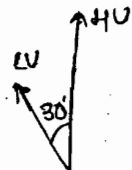
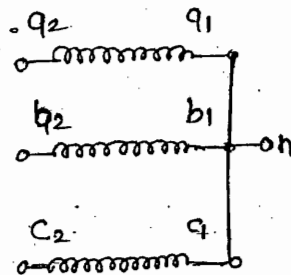
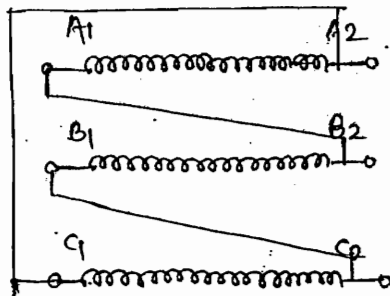
$S_{HV} = \sqrt{3}VI$

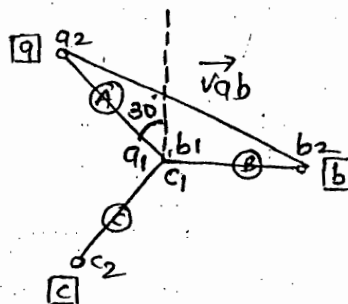
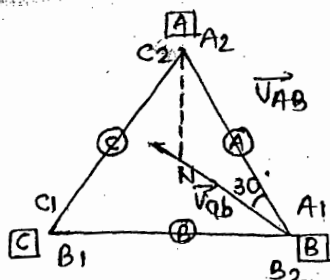
$S_{LV} = \sqrt{3} \left(\frac{\sqrt{3}V}{\sqrt{3}} \right) \left(\frac{q_1 I}{\sqrt{3}} \right)$

$= \sqrt{3}VI$

$S_{LV} = S_{HV}$

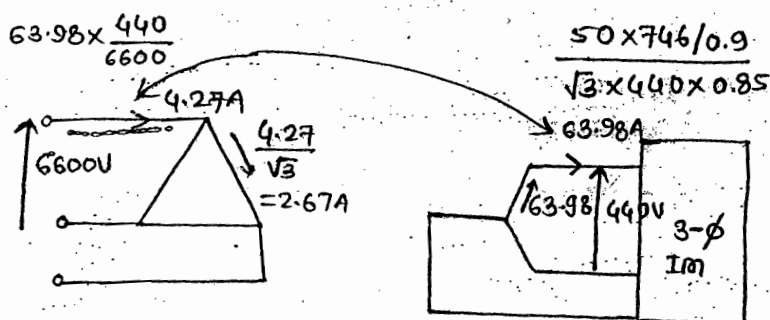
Dy11 →





Que. → A 50HP 440V 3 ϕ -Indⁿ motor with an η of 0.9 at a PF of 0.85 on FL is supplied from a 6600/440V Δ -Y connected Xmer. Ignoring the magnetising current cal. the currents in the HV & LV windings of TF when the motor is running at FL.

Solⁿ →



Que. → A λ - λ - Δ Xmer with 1^o, 2^o & 3^o voltages of 11kV, 1kV & 0.4kV has a magnetising current of 9A. There is a balance load of 600kVA at 0.8 PF lagging on the 2^o wdg & a balance load of 150kW on the 3^o wdg. Neglecting losses, find the 1^o & 3^o phase current if the 1^o PF is 0.82 lag.

Solⁿ →

$$I_1 N_1 = I_2 N_2 - I_3 N_3 - I_0 N_1$$

$$I_1 = \frac{I_2 N_2}{N_1} - \frac{I_3 N_3}{N_1} - I_0$$

$$\vec{S}_1 = \sqrt{3} \times 11 \times I_1 \angle \cos^{-1}(0.82) \text{ kVA}$$

$$\vec{S}_2 = 600 \angle \cos^{-1}(0.8) \text{ kVA} = 600 \angle 36.87^\circ \text{ kVA}$$

$$S_3 = \frac{150}{\cos \phi_3} \angle \phi_3 \text{ kVA} \quad (\text{assuming lag. PF } 3^\circ \text{ load})$$

$$S_0 = \sqrt{3} \times 11 \times 9 \angle 90^\circ \text{ kVA} \\ = 57.158 \angle 90^\circ \text{ kVA}$$

$$S_3 = \frac{150}{\cos \phi_3} \angle \phi_3$$

$$= \frac{150(\cos \phi_3 + j \sin \phi_3)}{\cos \phi_3}$$

$$S_3 = 150(1 + j \tan \phi_3)$$

$$S_1 = S_2 + S_3 + S_0$$

$$\sqrt{3} \times 11 \times I_1 \angle 0^\circ = 600 \angle \cos^{-1}(0.85) + (150 + j150 \tan \phi_3) + \sqrt{3} \times 11 \times 3 \angle 90^\circ$$

$$15.623 I_1 + j10.905 I_1 = 630 + j[360 + 150 \tan \phi_3 + 33\sqrt{3}]$$

equating real parts;

$$I_1 = \frac{630}{15.623} = 40.325 \text{ A}$$

equating imaginary parts;

$$10.905 = 360 + 150 \tan \phi_3 + 33\sqrt{3}$$

$$\phi_3 = 8.96^\circ$$

$$150 \times 10^3 = \sqrt{3} \times (0.4 \times 10^3) \times I_3(\text{line}) \times \cos(8.96^\circ)$$

$$I_3(\text{line}) = 218.945 \text{ A}$$

$$I_3(\text{ph.}) = \frac{218.945}{\sqrt{3}} = 126.41 \text{ A}$$

$$\boxed{I_1 = 40.32 \text{ A}, I_3 = 126.41 \text{ A}}$$

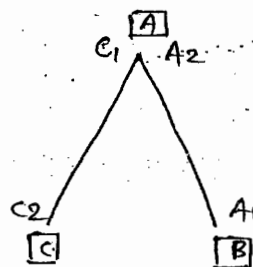
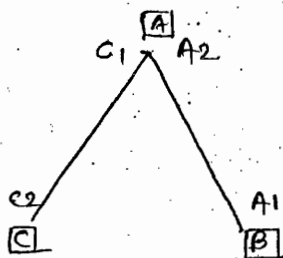
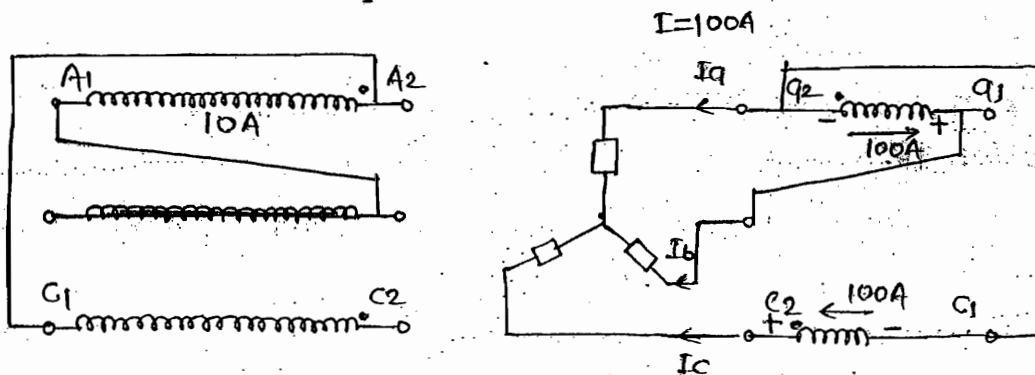
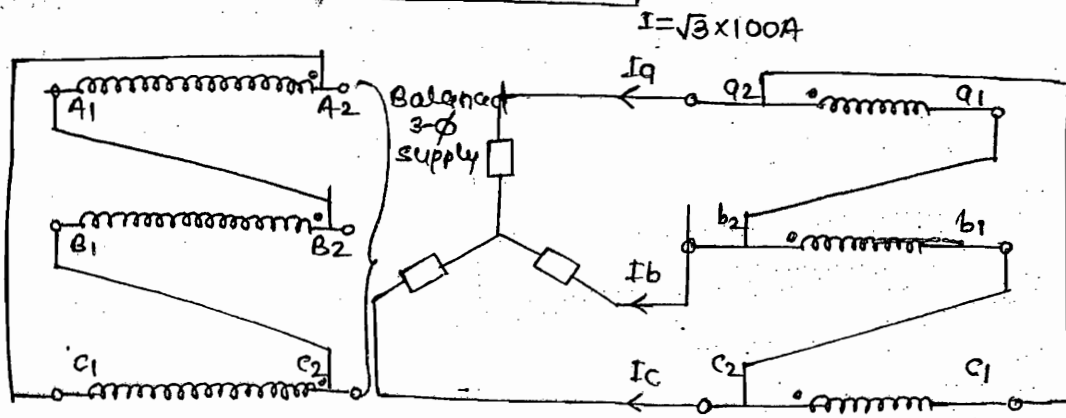
* star is used for HV, low capacity.

* Δ is used for LV, high capacity.

Applications of diff 3- ϕ TF connections $\rightarrow$

- * According to the general recommendation a λ connection is used for High vol., low capacity (1 kVA) applⁿ where as a Δ connection is Favoured for Low vol., High capacity applⁿ.
- * Since the kVA capacity of a TF is same on both sides, the natural choice would be a λ connection for HV side Δ connection for LV side.
- * Accordingly a λ - Δ connection is used for a step down applⁿ where- as a Δ - λ connection is used for step up applⁿ.
- * However there is an exception. In 2^o distribution sys. a neutral connection is req. to feed 1- ϕ loads as there are mixed ^(3 ϕ & 1 ϕ) loads connected to the system.
Therefore a Δ - λ connection is used in the step down mode in 2^o distribution TF.
- * A Δ - Δ connection is quite suitable to feed 3- ϕ loads of high capacity at LV levels.
If a 3- ϕ bank of 1- ϕ TF is used in Δ - Δ & if one TF has to be removed then the remaining 2 TF may still be used in open Δ (or V) connection to continue to supply 3- ϕ loads although at a reduced capacity of 57.7%.
- * Although a λ - λ connection appears to be quite attractive for HV applⁿ it is seldom used without a 2^o Δ wdg because of prob. related to magnetising current harmonics, unbalanced loads & unbalanced faults.

Vecor) Open Δ-connection



$$S_{\Delta-\Delta} = \sqrt{3} V \cdot I$$

$$S_{Vee} = \sqrt{3} \cdot V \left(\frac{I}{\sqrt{3}} \right)$$

$$= VI$$

$$\frac{S_{Vee}}{S_{\Delta\Delta}} = \frac{VI}{\sqrt{3} VI}$$

$$= \frac{1}{\sqrt{3}} = 0.577$$

$$\frac{S_{Vee}}{S_{\Delta\Delta}} = 57.7\%$$

$$S_L = \sqrt{3} VI \cos \phi$$

$$\vec{S}_C = VI \angle \phi + 30^\circ = \frac{SL}{\sqrt{3}} \angle \phi + 30^\circ$$

$$\vec{S}_A = VI \angle -(30^\circ - \phi)$$

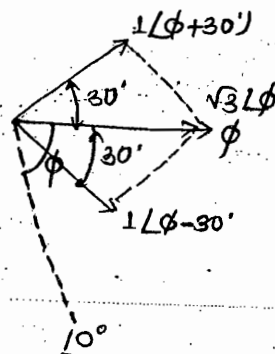
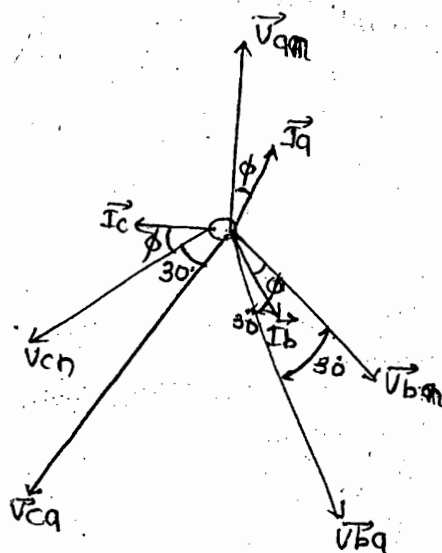
$$\vec{S}_A = VI \angle \phi - 30^\circ = \frac{SL}{\sqrt{3}} \angle \phi - 30^\circ$$

$$\vec{S}_C + \vec{S}_A = VI [1 \angle \phi + 30^\circ + 1 \angle \phi - 30^\circ]$$

$$= VI [\sqrt{3} \angle \phi]$$

$$= \sqrt{3} VI \angle \phi$$

$$\boxed{S_C + S_A = \vec{S}_{load}}$$



Que. → A 3ϕ 1000 kVA 0.866 lag PF load is supplied by V connection at 400V. Determine the kVA o/p & operating PF of each TF. Neglect exciting current & all losses.

Soln. →

$$\phi = \cos^{-1}(0.866)$$

$$= 30^\circ \text{ lag}$$

$$\vec{S}_C = \frac{1000}{\sqrt{3}} \angle 30^\circ + 30^\circ$$

$$= 0.577 \angle 60^\circ \text{ kVA}$$

at PF 0.5 lag.

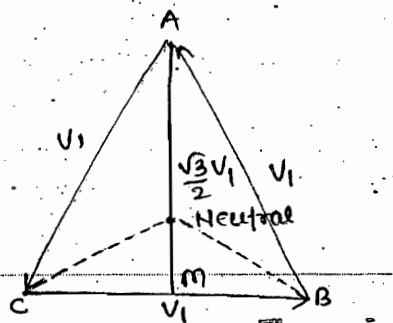
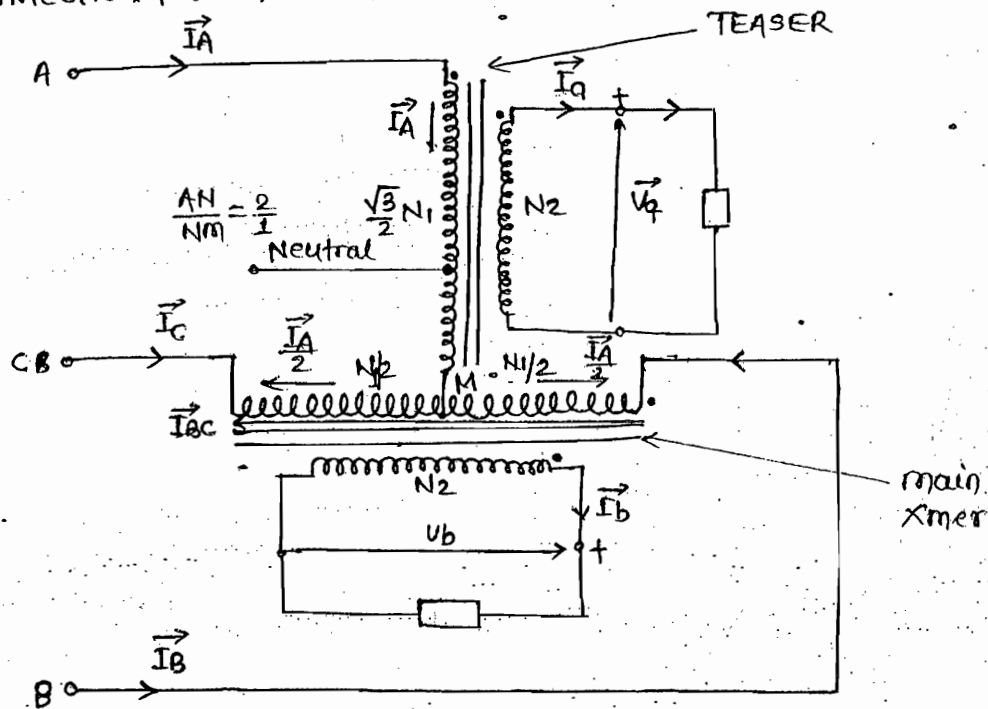
$$\vec{S}_A = \frac{1000}{\sqrt{3}} \angle 30^\circ - 30^\circ$$

$$= 0.577 \angle 0^\circ$$

at PF = 0.

SCOTT CONNECTION

* Scott connection for 3- ϕ to 2- ϕ conversion.



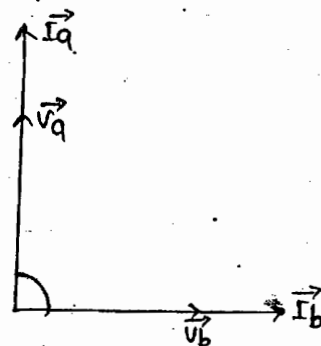
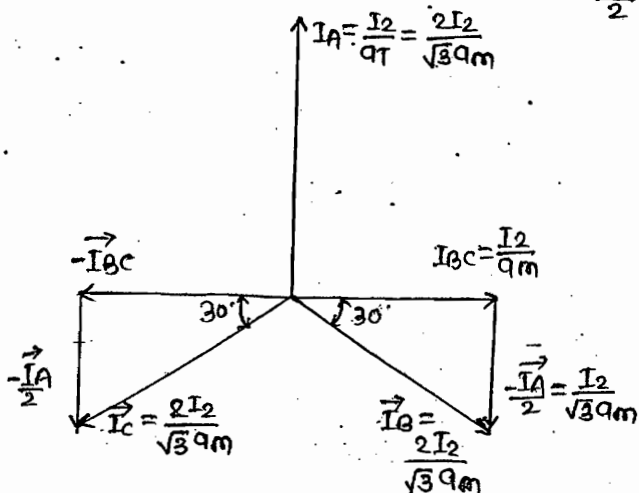
$$q_m = \frac{N_1}{N_2}$$

$$q_T = \frac{\sqrt{3} N_1 / 2}{N_2} = \frac{\sqrt{3}}{2} q_m$$

$$V_b = \frac{V_1}{q_m} ; V_a = \frac{\sqrt{3} V_1 / 2}{q_T} = \frac{\sqrt{3} V_1 / 2}{\frac{\sqrt{3}}{2} q_m / 2}$$

$$V_a = \frac{V_1}{q_m} = V_b$$

$$\vec{I}_B = \vec{I}_{Bc} - \frac{\vec{I}_A}{2} ; I_c = -I_{Bc} - \frac{I_A}{2}$$



unity pf load. ($I_a = I_b = I_2$)

$$S_{\text{main}} = V_L I_L \quad S_{\text{teaser}} = \frac{\sqrt{3}}{2} V_L \times I_L$$

$$\frac{S_{\text{main}}}{S_{\text{teaser}}} = \frac{2}{\sqrt{3}} = 1.15$$

$$\boxed{\frac{S_m}{S_T} = 1.15}$$

Que. → 2- ϕ furnaces A & B are supplied at 80V by means of Scott connected Xmer^{combination} for a 3- ϕ 6600V system. The vol. of Furnace A is leading. Cal. the line currents on the 3- ϕ side when

(a) The furnaces take 800kW each at 0.8 PF lag.

(b) Furnace A takes 500kW at UPF & B takes 800kW at 0.707 PF lag.

Soln → let; $V_B = 80 \angle 0^\circ$, $V_A = 80 \angle 90^\circ$

(a) $a_m = \frac{N_1}{N_2} = \frac{6600}{80}$

$$a_m = 82.5 \quad a_T = \frac{\sqrt{3}}{2} a_m = \frac{\sqrt{3}}{2} \times 82.5 = 71.45$$

$$a_m = 82.5, a_T = 71.45$$

$$I_b = \frac{800 \times 10^3}{80 \times 0.8} \angle -\cos^{-1}(0.8) = 12500 \angle -36.87^\circ A$$

$$I_a = \frac{800 \times 10^3}{80 \times 0.8} \angle 90 - \cos^{-1}(0.8)$$

$$I_a = 12500 \angle 53.13^\circ A$$

$$\vec{I}_{BC} = \frac{\vec{I}_b}{a_m} = 151 \angle -36.87^\circ A, \quad \vec{I}_A = \frac{I_a}{a_T} = 174.95 \angle 53.13^\circ A$$

$$\vec{I}_B = \vec{I}_{BC} - \frac{\vec{I}_A}{2}$$

$$= 174.96 \angle -66.87^\circ A$$

$$I_C = -I_{BC} - \frac{I_A}{2}$$

$$= 174.96 \angle 173.13^\circ A$$

$$\boxed{I_A = 174.95 \angle 53.13^\circ A, I_B = 174.96 \angle -66.87^\circ A}$$

$$\text{--- } I_C = 174.96 \angle 173.13^\circ A$$

(b)

$$\vec{I}_b = \frac{800 \times 10^3}{80 \times 0.707} \angle -\cos^{-1}(0.707)$$

$$= 1414.427 \angle -45^\circ \text{ A}$$

$$\vec{I}_a = \frac{500 \times 10^3}{80 \times 1.0} \angle 90^\circ$$

$$= 6250 \angle 90^\circ \text{ A}$$

$$\vec{I}_{bc} = \frac{\vec{I}_{ab}}{q_m} = 171.45 \angle -45^\circ \text{ A}$$

$$\vec{I}_a = \frac{\vec{I}_a}{q_T} = 87.47 \angle 90^\circ \text{ A}$$

$$\vec{I}_B = \vec{I}_{bc} - \frac{\vec{I}_a}{2} = 204.72 \angle -53.69^\circ \text{ A}$$

$$\vec{I}_c = -\vec{I}_{bc} - \frac{\vec{I}_a}{2} = 143.89 \angle 147.47^\circ \text{ A}$$